

ANALYTICAL METHODOLOGIES FOR ELECTROACOUSTIC MUSIC: A SEMIOTIC CLASSIFICATION SCHEME AND CRITICAL SURVEY

Rodrigo F. Cádiz
rcadiz@uc.cl

Music Institute and Department of Electrical Engineering
Pontificia Universidad Católica de Chile, Chile

Miriam Akkermann

Institut für Musik und Musikwissenschaft, Technische Universität Dortmund, Germany

Daniela Fugellie

Instituto de Música y Centro de Estudios Mediales, Universidad Alberto Hurtado, Chile
Federico Eisner-Sagüés

Facultad de Artes, Pontificia Universidad Católica de Chile

Electroacoustic music has not only expanded the horizons of instrumental music by delving into a broader realm of sonic possibilities based on technological sound processing and generation, but has also given rise to a novel artistic expression. The absence of a conventional written score coupled with the relatively recent emergence of electroacoustic music introduces distinctive challenges to its analytical exploration. Consequently, a unified analytical lexicon, tailored to the unique features of this genre, is yet to be established. To address this gap, we outline foundational considerations for approaches and subsequently embark on a comprehensive survey and critical evaluation of key analytical methodologies proposed in the literature. Employing a triangular semiotic framework, we classify each analysis technique based on its relationship to the poietic, aesthetic, and trace axes of the triangle. Our central contribution is this classification scheme, which positions every reviewed methodology within a single triangular space and supplies the organizing thread of the survey. The systematic review and classification serve as valuable tools in fostering the development of a cohesive analytical lexicon, providing robust groundwork for the future evolution of comprehensive methodologies in the analysis of electroacoustic music.

1. INTRODUCTION

Electroacoustic music, defined as music that is generated by electronic means, or by a combination of these with acoustic instruments, constitutes a very particular genre in the history of music. This music manifests important differences with respect to instrumental music, among them the usual absence of a written score and known sound sources, and the possibility of composing not only between sounds, but within them (Stockhausen 1998). While some composers encompass most sound arts under the term music (Landy 2007), others have questioned if what they do can still be denominated music in the traditional sense of the word, and reject the notion of being a composer (Morthenson 1985). In general, these artists understand music as a subset of a more general sonic art, a new art form that, as Jean-Claude Risset states, is so different from instrumental music that it belongs to a totally different branch, in the same way that cinema is different from theatre (Risset 2002). It has been suggested that electroacoustic music is a meta-aural medium, a kind of “meta-music” (Camilleri and Smalley 1998). A number of scholars have addressed the analysis of electroacoustic music specifically, among them Camilleri and Smalley (1998), Battier and Landy (2004), Mahnkopf (2006), Simoni (2006) and, more recently, Emmerson and Landy (2016); their contributions are reviewed in detail in the sections that follow.

The distinctive characteristics of electroacoustic music significantly impact the analysis process. Unlike instrumental music, many conventional analytical methods are not directly applicable, leading to the development of various strategies and methodologies in the literature. Each proposed approach comes with its own set of strengths and weaknesses, yet there remains a lack of overarching strategies, methods, or an established analytical lexicon—a crucial element for comprehensive analysis. Typically, the analysis of electroacoustic music is conducted heuristically,

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tailored to the unique attributes of each individual work. Despite the widespread use of representation of specific measures such as the sonogram, there is a notable absence of universally accepted strategies, underscoring the complexity and idiosyncrasies in electroacoustic music. Recognizing this inherent complexity becomes particularly vital in navigating the nuanced landscape of electroacoustic music analysis.

Electroacoustic music is a perfect example of a genre that challenges the field of analysis, given the intricate nature of its composition, inherent constraints, sustained timbral dimension, distinctive electronic instrumentation, spatial considerations, and the intricacies of its performance. Recognizing these complexities, a thorough examination of the unique characteristics inherent in this genre becomes indispensable, as emphasized by Bossis (2006).

1.1. Related work

Since 2000, several authors have attempted to find invariant features and common strategies, hoping for a systematization of electroacoustic music analysis. One compilation of analyses is provided in the book by Licata (2002), which contains analysis of seminal works by composers such as Lucier or Stockhausen. The volume gathers diverse analytical perspectives—from pitch- and set-based readings to spectromorphological and perceptual approaches—applied to canonical electroacoustic works.

A year later Roy (2003) published a book where he reviews, criticizes, and proposes different methodologies for the analysis of electroacoustic music, and applies several type of analysis to the same piece. Roy's approach is based on Nattiez's semiology of music (Nattiez 1990), a foundation that we also adopt, and think to be appropriate for a classification scheme for the different electroacoustic music analysis strategies that have been proposed so far. Roy's book presents various methods applied to iconic works of electroacoustic music and five approaches borrowed from acoustic music that might be applicable to electroacoustic music. These are the paradigmatic analysis by Ruwet, the generative analysis by Lerdahl and Jackendoff, the implicative methodology by Meyer, and Roy's functional analysis (Zattra 2005).

Bossis (2006) proposes to establish the terms and conditions of a systematic musicology of electroacoustic music. The systematization of research allows invariants to be revealed through rigorous documentation and a coherent analytic corpus. These common elements are useful to a reasoned vision of styles, genres, and a topology of electroacoustic music. Many in-depth analyses and historical studies follow rigorous procedures, such as questioning the status of sources, and understanding their importance can ease the understanding of this heritage. In the same way that ancient Western music has been re-categorised according to time periods, electroacoustic music can be included in different bodies of work.

Zattra (2005) reviews several analytical approaches, with a strong focus on computational strategies, as an attempt to organize the knowledge and competence that the electroacoustic music community has reached in analysing its product until that point in time. Zattra classifies the analysis into two main categories: analysis by musicologists and computational analysis, clearly establishing a division between analysis done by a human observer and by a machine.

Simoni (2006) edited a book that contains analysis of some important works that illustrate different approaches and methodologies, such as musical signals visualization, the intention/reception project, computational analysis and the role of spatialization in electroacoustic music.

Lalitte (2006) developed a semiotic model for the analysis of mixed music, where a semiotic square diagram is used for discovering meaningful relationships between the acoustics and electronics parts of the music. This type of analysis is used to confront two terms and their negatives, in this case a real, a virtual source, real-time processing and sound projection. Zattra (2004) has also proposed a semiotic-based model, based on two poles. In one extreme, there is the aesthesical

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analysis, approximating the music from the point of view of perception, while in the other dwells the poetical analysis, which studies the creative process.

A very important recent effort is the book by Emmerson and Landy (2016), which provides a state-of-the-art overview of the most advanced analysis methods for electroacoustic music, with a special care of differentiating electroacoustic genres and their particularities. Perhaps the most important contribution of this work is, as the book title suggests, the expansion of the analysis to other genres usually not associated with the acousmatic, such as computer games, live and interactive music, turntable music and sound art.

A special mention deserves the OREMA project (Gatt 2015, 2016) which aims to generate an open access knowledge repository and a platform that will facilitate the interchange between creators of original content and its audience. This is very important for the analysis of electroacoustic music, as open access is essential for the establishment of an analytical community. The geographic and institutional spread of such activity is illustrated by initiatives like the Hong Kong SoundLab (Ikeshiro and Lindborg 2021).

The principal contribution of this article is a semiotic classification scheme that arranges the existing analytical methodologies according to the tripartite model of Molino and Nattiez, and the review that follows is structured around it. More precisely, the article pursues two goals: first, to introduce to the field a systematic and novel organisation of the many analytical approaches that have been proposed for electroacoustic music; and second, to represent this complex landscape concretely—both as a map that situates each method, together with its principal authors, in relation to the three semiotic levels (Figure 2 and Table 1), and as the classification scheme that we propose. This article is structured as follows. First, general considerations for the analysis of electroacoustic music are discussed including the establishment of an analytical lexicon and categorization. Second, a critical and exhaustive review of many and different methodologies for analysis is presented, classified by their degree of relation to each pole of the tripartite model of Molino and Nattiez, detailing the most important aspects and difficulties of each proposal. Third, poetic, aesthetic, neutral level and mixed analyses are reviewed and discussed. Finally, our main conclusions are presented.

2. GENERAL CONSIDERATIONS FOR THE ANALYSIS OF ELECTROACOUSTIC MUSIC

The analysis of electroacoustic music is a complex and heterogeneous discipline. Electroacoustic music subgenres and styles vary greatly, and the collection of instruments, devices and sound processing techniques is also very heterogeneous, making the analysis very difficult (Zattra 2005, Bossis 2006).

A very important factor in the analysis of electroacoustic music is the usual lack of a visual representation of the music (that is, of a score-like notation) (Licata 2002). As a way of overcoming this problem, analysts often use visual strategies such as sonograms or composer's personal notes of the work being studied. Also, as Gayou (2006) emphasizes, analysts often use inadequate words for describing audio and music phenomena. It is common that on the first glance, different authors write in different terms but, when studied in detail and depth, they are referring to the same issue (namely, the lack of a shared analytical vocabulary). These two obstacles are a clear indication that there is a lack of an exhaustive and consistent analytical lexicon for electroacoustic music, as Camilleri stated almost twenty-five years ago (Camilleri 1993). There is still not a universally accepted methodology for the analysis of electroacoustic music, which results in an urgent need for general strategies and methodologies for analysis. Bossis (2006) proposes to search for invariants, common methodologies that would allow for the systematization of analysis. A recurring premise of this article is therefore that analytical methods cannot simply be imported from instrumental music: they must be adapted to the defining features of the electroacoustic genre—the usual absence of a score, the centrality of timbre and texture, the constitutive role of space and diffusion, and the dependence of the work on its technological means of production. Each of the

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methodologies reviewed below can be read as a different way of accommodating one or more of these features.

The analysis of electroacoustic music is generally done on a case by case basis and it is very particular to the piece being analysed. Although this is generally true for most music, in the case of this particular genre, the complexity and amplitude of its domain make this situation even more relevant. Probably the most indicative fact of the problems and challenges that the analysis of electroacoustic music faces today is that in spite of some notable advances in the last few years (Emmerson and Landy 2016), published analyses of electroacoustic music are still very rare in comparison to analytical publications of instrumental or even contemporary music, a disparity that also reflects the far larger instrumental repertoire accumulated over several centuries.

What is the use of music analysis? What do we analyse? What does musical analysis tell us when the object is a fixed sound recording rather than a notated score? Analysis, as theory and praxis, has undergone a period of self-questioning, no doubt resulting from its inability to account for the new musical techniques championed by the post-war avant-garde (Norris 1999). Even today, much contemporary music is seemingly analytically inscrutable. This is evident in the context of electroacoustic music. Akkermann (2025) revisits this question from a present-day vantage point, relating the changing musical needs of composers and analysts to the intelligent tools now available. This renewed interest in analysis is not confined to electroacoustic music: in the Spanish-speaking world it is reflected in the recent creation of the Sociedad de Análisis y Teoría Musical (SATMUS) and its journal *Súmula* (SATMUS 2023).

Camilleri (1993) stated twenty-five years ago that the introduction of technology in its various phases of development from the end of the 1940's had brought about not only an enrichment of the sound palette, but it had also brought a great deal of theoretical reflection on how to classify the works thus created and how to analyse them.

What is the objective of analyzing electroacoustic music? The response to this question varies greatly depending on the author. Some, such as Delalande (1998), emphasizes that the object of analysis is the sound. Others, such as Zattra (2005) recognize the material as the nature of music difficult to define, implying that the object of analysis must be clearly defined, and it can be as varied as the score or sound image, the mental image of the composer or its performance. This is where a semiotic model can be of use for the analyst.

Returning to the question of what analysis is for, some concrete answers have been proposed, even if they have not reached a broad audience. For example, Camilleri (1993) proposed that the creation of analytical strategies for electroacoustic music must regard three aspects:

1. A reflection on the musical language, its potentialities and its ties with the natural sound world.
2. The relationship between psycho-acoustic properties of sound phenomena and their description.
3. The creation of an analytical lexicon.

These aspects are essential and equally important, because they allow creating an appropriate framework to the analysis process. But the third is especially important, in our view, due to the general lack of general and common analysis methods and strategies for electroacoustic music, in contrast to more traditional forms of analysis. An important goal of any analytic method is therefore to provide a basic framework or lexicon that is consistent and exhaustive. A recent pedagogical contribution in this direction is Kaiser (2025), who develops a framework for the analysis of contemporary electronic music.

We feel that the electroacoustic music community needs now to start organizing all the different approaches that have been proposed in the literature, in order to exploit their results for the creation of common analysis approaches. As Zattra (2005) proposes, it is time to reflect on all

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methodologies, by evaluating their merits and defects, and by proposing possible ways of mixing them. This is what drives our critical review. As Camilleri and Smalley wrote: “If we can ‘understand’ our relationship to the wide-ranging sound-world of electroacoustic music, then we shall be better positioned to arrive at a more comprehensive understanding of music and listening as cultural practices: that is the longer-term ambition of an analytical agenda centred on electroacoustic music” (Camilleri and Smalley 1998).

Before turning to specific methods, it is worth making explicit where the categories used in the following subsections come from. Any analysis of electroacoustic music, whatever its semiotic orientation, must come to terms with a small set of cross-cutting dimensions: how sonic events are delimited (Section 2.1), how movement and motion are construed (Section 2.2), how form is articulated (Section 2.3), and how space is organised (Section 2.4). These dimensions are not themselves analytical methods but the questions that every method must answer, and they recur throughout the methodologies surveyed in Sections 4 to 7.

2.1. Events

The concept of event is essential for most of the analysis methods of electroacoustic music, and it is something that deserves some thought. At the moment of analysis, the dimension being studied needs to be clear (Zattra 2006), as well as the precise form events are determined and considered. The account that follows draws in particular on Schaeffer’s sound-object theory and on Kendall (2008) and Aguilar (2006).

The concept of event is an understudied notion and it is generally assumed that an event is something clearly delimited and free from any confusion and that the recognition of an event is automatic (Kendall 2008). According to Schaeffer, all objects are perceived in terms of their containing structure, and structures are perceived in terms of the objects they contain (Schaeffer 1966). A perceptual object that is constructed based on other smaller objects, can be at the same time considered as a unit belonging to a structure, or as a structure (Aguilar 2006). This is closely related to the concept of an event that can be perceived as such, or as a part of a bigger event, that operates at another level of hierarchy.

In traditional music, this dualism is very clear. In electroacoustic music, this is more difficult to establish (Aguilar 2006). Some computational analysis approaches, such as SQEMA (Park 2016), include an “event level analysis”, whose aim is to analyse the identified events using a clear nomenclature. In this respect, Kendall (2008) proposes an event schema, a dynamic model that includes parts representing processes and other states, that can re-adapt itself as time progresses. The importance of this schema is that when analysing electroacoustic music, it is important to adequately delimitate and characterize events, and not to assume them as something defined and established a priori.

2.2. Movement

Movement in electro-acoustic music deserves a special treatment (Kendall 2006). Here, by ‘movement’ we mean the perceived sense of motion in the music—whether evoked by spatialisation or merely implied by spectral and dynamic shaping—rather than physical displacement alone. In general, most kinds of analysis assume that music is movement and it is assumed that the analysis must account for the relationships between musical elements. It is usually taken for granted that music is motion in time. The evolution of electroacoustic music has made us progressively more aware of an extensive range of types and patterns of motion. Musical motion has been made even more apparent through the advent of state of the art multi-channel diffusion systems. Noble, Bonin and McAdams (2020) extend the discussion of musical time, examining experiences of time and timelessness specific to electroacoustic music.

However, as Smalley (1986) emphasizes, spectro-morphological design on its own, by means of spectral and dynamic shaping, is able to create real and imagined motions without the need for

actual movement in space (i.e. spatialisation, as distinct from the implied motion defined above). This type of motion may apply at a variety of structural levels and time-scales, from the shape of a brief sound-object to the motion of a large structure, and from the groupings of objects to the groupings of much larger structures.

Kendall (2006) shows that in certain music, non-movement is a fundamental concept, such as the case of Varèse's *Poème Electronique*. This concept is closely associated with the phenomenon of juxtaposition, understood not only as a product of perceptual processes such as audible streams (Bregman 1990) or simply contrast. Juxtaposition in this situation implies a divorce or disjunction between otherwise unrelated sound materials, not only a difference.

Cook (1998) understands the analysis as a compromise with the idea of perceiving the music as a series of interacting elements. Juxtaposition, on the contrary, tries to position different elements without establishing connections of relations between them. Juxtaposition implies no directionality. When a gamma of very different material is presented in a composition, something very common in electroacoustic music, juxtaposition is relatively easy to achieve. Juxtaposition thus bears directly on movement: by denying directionality between elements, it withholds the sense of motion that connected, interacting materials would otherwise create.

Movement can also manifest different kinds of perspectives. Truax (2008) observes two structural strategies when composing, particularly for soundscapes, one that rely on a fixed spatial perspective, and another that rely on moving perspectives. A fixed perspective implies a continuity of space, with a temporal flow constituted by sonic events. On the opposite, a temporal perspective relies on spatial transitions, usually simulated or evoked. This creates a sense of travel and narrative.

In Smalley's proposal (Smalley 1986), motion is classified into six categories (bi-directional, uni-directional, linear, curved-linear, reciprocal and centric/cyclic), correlated among themselves by diverse degrees of affinity and subdivided into subcategories. This definition of categories of motion is of great importance since these categories can be related to single sound objects as well as to sections of a work. Smalley indicates this difference with the terms texture and gesture, to indicate whether these categories refer to the internal structure of a sonic object or whether the reference is to a movement of greater breadth, a section of a work.

2.3. Form

A possible definition of form in music is a way of thinking about how we engage with a sense of connectedness and completeness in a work (Young 2016).

Young (2016) differentiates various aspects of form in electroacoustic music, denoted as musical rhetoric, the idea of 'moment' form, the idea of 'morphic' form and the idea of narrative.

Musical rhetoric concerns the way a work directs attention through contrast and emphasis; 'moment' form, after Stockhausen, treats sections as self-contained presents rather than goal-directed progressions; 'morphic' form foregrounds the growth and transformation of sound-shapes; and the idea of narrative organises the work as a sequence of events from which the listener constructs meaning over time.

2.4. Space

Space, like movement, deserves to be defined at the outset: the term covers both the disposition of sounds within a real or virtual acoustic image and their displacement across a multichannel reproduction system. In most electroacoustic music, space is a fundamental concept, especially in the case of multichannel pieces. Even stereo pieces are usually spatialized live by composers or performers. Otondo (2008) compared the information gathered by conducting a spatialisation

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survey in 2008 with seventeen articles by composers written on the topic in 1997, finding that composers nowadays are more used to working with different types of spatialisation systems than previously.

Listeners experience electroacoustic music as full of significance and meaning, and spatiality is one of the contributing factors to its meaningfulness (Kendall 2010). The development of progressively more refined technological tools for handling space has had an impact on many of the different features of electroacoustic music, including spatialisation (Otondo 2008). For analysis purposes, it is important to consider how composers can transform spatial information to satisfy the requirements of the structural development of a piece, but also to first consider the importance of the spatial awareness of the listeners (Barrett 2002).

Kendall and Ardila (2007) noticed a lack of a definitive and accepted vocabulary when referring to space, not only when referring to the spatial attributes of individual sounds, but also to describe how these attributes participate in the artistic expression. In order to solve this problem, Kendall (2010) has proposed an extension of a previous vocabulary spatial framework by Rumsey, consisting of dimensional attributes (Direction, Distance, Extent: depth, width and height) and Immersive attributes (Presence, Envelopment). This vocabulary is a valuable tool for the analyst interested in the spatial attributes of a piece of music. Merlier (2018) addresses the complementary problem of notating space, proposing a system of signs to represent spatial gestures.

Barrett (2002) emphasizes that “although we normally perceive the sound of the object and the space as a totality, for purposes of composition and analysis it is useful to separate the two”. For Kendall (2010), there is a distinctive and intricate linkage between meaning and space, as spatial meaning emerges from the idiosyncratic characteristics of multichannel reproduction, something he refers to as “the nature of the artistic medium itself”.

Barrett (2002) and Kendall (2010), for example, show how the analytical separation of sound and space can be turned into an explicit parameter when studying multichannel works.

3. A SEMIOTIC-BASED CLASSIFICATION SCHEME

Semiotics can be defined as the science or study of signs. In this article, we prefer to use the wider term semiotics instead of the term semiology, which is usually associated with linguistics, i.e. the study of signs as if they were words.

3.1. Semiotic analysis

Semiotic analysis, as proposed by Molino and applied to music by Nattiez (1990), consists of three levels of description: the poetic, the trace or neutral level, and the aesthetic. The underlying assumption of this analysis is that music is no longer the “text” or the “structure”, but it is also constituted by the procedures that have given birth to it and the effects that it generates as acts of perception. Poetics describes the *link* among the composer's intentions, his creative procedures, his mental schemas, and the *result* of this collection of strategies; that is, the components that go into the work's material embodiment. Poietic description deals, among others, with a special form of hearing: what the composer hears while imagining the work's sonorous results, or while experimenting at the piano, or with tape. This was called by Varèse the interior ear. Opposite to poietics, aesthetics is not merely the artificially attentive hearing of a musicologist, but the *description of perceptive behaviours* within a given population of listeners; that is how this or that aspect of sonorous reality is captured by their perceptive strategies. The neutral level is that of the immanent physical *trace*, the score in the case of instrumental music or the sound-object in the case of electroacoustic music. The neutral level is thus created and interpreted by the aesthetical level and the poietical level. The tripartite model remains the most widely used framework in music semiotics, and scholarship from 2016 to 2026 has continued to apply, extend and contest it—

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integrating it with Peircean and cognitive-semiotic approaches (Curry 2017; Reybrouck 2017; Giacosa 2023) and debating the status of the neutral level (Harcus 2017; Nattiez 2019; Pihel 2024). We adopt it here not as a settled doctrine but as a robust, well-understood scaffolding on which the diverse strategies for analysing electroacoustic music can be located and compared. Beyond these debates, the model has been hybridised with Peircean semiotics (Curry 2017; Alvear et al. 2024), with Tarasti's existential semiotics (Tarasti n.d.; Bramantyo 2025) and with cognitive and embodied approaches (Reybrouck 2017; Giacosa 2023; Postiga et al. 2025), and it remains a reference point for the semiotic study of timbre and sound-symbolic form (Sad 2020); although these developments have largely been pursued outside electroacoustic music, they bear directly on the analytical questions raised here.

These three levels form a tripartite semiotic scheme, as shown in figure 1:

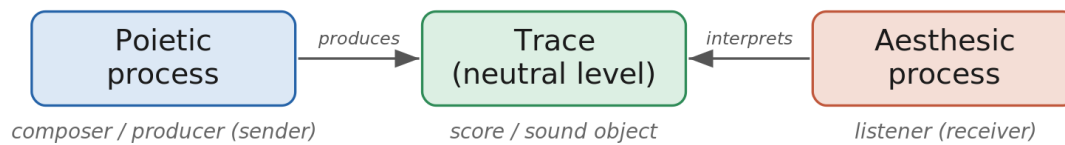


Figure 1. Molino's tripartite model of semiotic analysis. Both the poietic and aesthetic processes affect the trace or neutral level. The trace is created and interpreted by the poietical and aesthetical levels.

The tripartite also resonates with broader semiotic models of communication, such as Eco's sender–message–receiver triad, in which meaning is actively negotiated rather than simply transmitted. Semiotic analysis proposes that, by examining these three levels in turn, it is possible to grasp music in all its multifaceted totality. Following Nattiez (1990, 15), two observations are important. First, as figure 1 shows, the poietic and the aesthetic are processes, where immanent configurations are organised into structures. And second, it is important not to confuse the trace with the analysis of that trace. The trace is merely a physical reality without shape, until it is “entrapped” by analysis. It must be stressed, however, that the trace, or so-called neutral level, is never genuinely neutral: the choice of representation and measurement already embodies analytical decisions, which is why some authors prefer to speak of a ‘source’ rather than a ‘neutral’ level.

3.2. Classification scheme

We propose a classification scheme based on the semiology of music, where each methodology is categorized according to their function in terms of the tripartite proposed by Molino and Nattiez (1990, 17). This is shown in figure 2, where each analysis technique considered in this review is classified according to their degree of relation to one of the three semiotic vertices in a triangle: poietic, neutral level or aesthetic. We have decided to put the levels in the extremes of a triangle in order to be able to position each type of analysis in a two-dimensional space, where the distance to the vertices can be interpreted as how each technique relates to each of the semiotic levels.

In some aspects, this is an extension of Zattra's analysis model (Zattra 2004) - which aims to combine poietic and aesthetic analysis in order to understand the relationship between technology and the actual piece of music - by incorporating the trace or neutral level. We are fully aware that this classification scheme is neither absolute nor perfect, and that there are blurred boundaries between methodologies and that some of them do not completely fit into this tripartite model — though this is not in itself a problem, since the scheme is intended as an orientation from which each analyst may take whatever is useful for a given work, rather than a set of rigid categories. It is also possible that the reader would disagree on the placement of a particular analysis

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methodology in the overall scheme. However, we hope that this classification scheme could be refined in time by other authors and adopted towards an establishment of an analytical lexicon as desired by Camilleri (1993). A semiological reading in this spirit is offered by Salynas (2021), who interprets the electroacoustic phenomenon in explicitly semiotic terms.

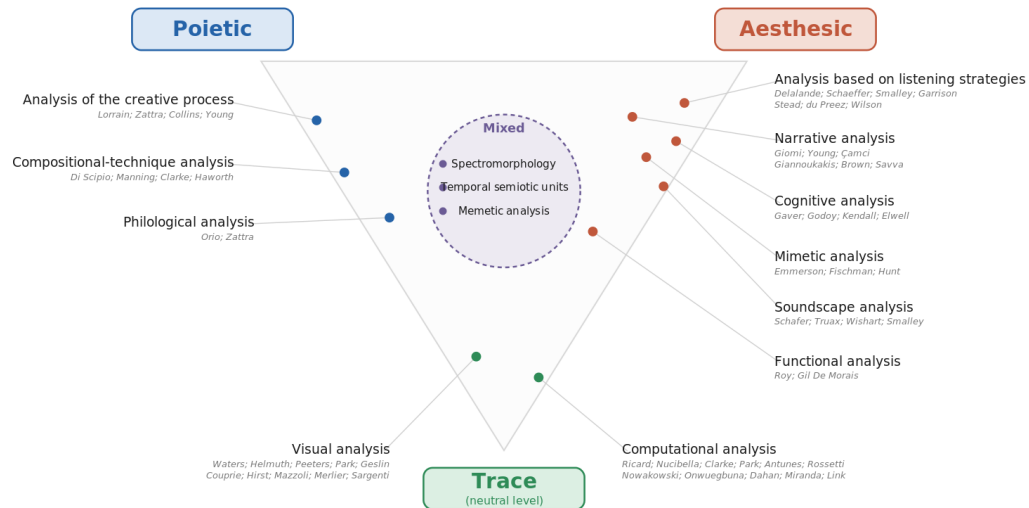


Figure 2: A classification scheme of analytical methodologies based on the tripartite of semiotic analysis. Methodologies are classified according to their semiotic nature on a three axes space. Each family is placed near the semiotic axis (or axes) to which it relates most closely, while the genuinely mixed approaches are gathered in the central circle. Figures 2 and 3 group the analytical approaches into families (the categories of Sections 4 to 7); the individual methods that make up each family are listed in Table 1.

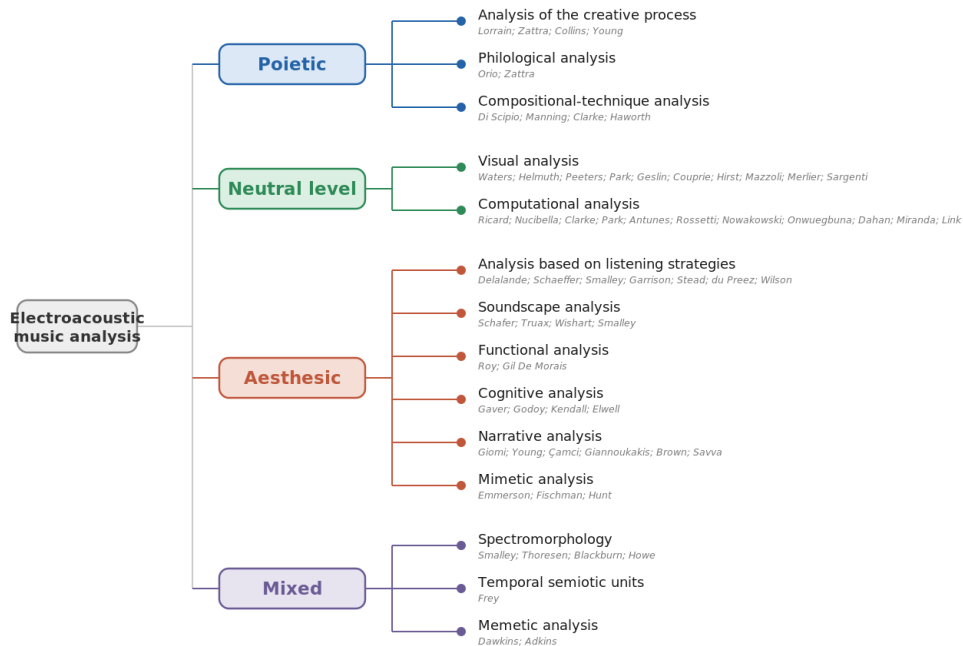


Figure 3. The same classification represented as a dendrogram. The families of analysis methods are grouped by their dominant semiotic level (poietic, neutral, aesthetic or mixed); the names beneath each method are the principal authors of the works discussed in the corresponding sections.

Table 1. The analysis techniques reviewed in this article and their position within the tripartite scheme. Each technique corresponds to one mapped in Figure 2.

Analysis technique (family)	Semiotic level	Specific methods and representative references
Analysis of the creative process	Poietic	creative-process / generative analysis (Lorrain 1980; Zattra 2005); analysis of generative programs (Collins 2008; Young 2016)
Philological analysis	Poietic	Orio & Zattra 2007; Zattra 2007
Compositional-technique (techné) analysis	Poietic	techné analysis (Di Scipio 1995, 1998; Manning 2006); technological investigation & software emulation (Clarke 2014); synthesis-procedures-as-texts (Haworth 2015)
Visual analysis	Neutral level	sonogram / spectrogram (Waters & Ungvary 1990; Helmuth 1996); timbregrams (Park 2016); rhythmograms (Hirst 2014); self-similarity (Peeters et al. 2002); aural transcription (Mazzoli 2014, 2017); space notation (Merlier 2018); annotation environments — Acousmographie, EAnalysis (Geslin & Lefevre 2004; Couprie 2016)
Computational analysis	Neutral level	audio descriptors / MIR (Park 2016; Antunes et al. 2019; Rossetti et al. 2018, 2019); morphological sound description (Ricard & Herrera 2004); multidisciplinary analysis (Nucibella et al. 2005); interactive aural analysis / TIAALS (Clarke 2012; Clarke et al. 2013); deep-learning transcription (Nowakowski et al. 2019; Onwuegbuna 2019); comparative analysis (Cassaro 2018); temporal framework (Dahan 2011, 2020); neurotechnology (Miranda 2010); immediate constituent analysis (Link 2018)

Analysis based on listening strategies	Aesthetic	reception behaviours (Delalande 1998); Schaefferian modes & surrogacy (Smalley 1992); modes of listening (Wilson 2023); intention/reception (Stead 2009); model listener (Garrison 2013; du Preez 2022); improvisation analysis (Couprie 2022; Hyvärinen 2022)
Soundscape analysis	Aesthetic	Schafer; Wishart 1986; Truax 2001, 2008
Functional analysis	Aesthetic (neutral aspects)	Roy 1998, 2003; Gil De Moraes 2015; integrated / multimodal acousmatic analysis (Couprie et al. 2025)
Cognitive analysis	Aesthetic	mental layers (Kendall 2014, 2016); embodied cognition (Godoy 2006); everyday listening (Gaver 1993); aural-space taxonomy (Elwell 2023)
Narrative analysis	Aesthetic	narrative strategies (Giomi & Ligabue 1998; Young 2016; Çamci 2016; Giannoukakis 2016); dramatized-performative narrative (Savva 2021)
Mimetic analysis	Aesthetic	Emmerson 1986; Fischman 2008; voice and text (Hunt 2020)
Spectromorphological analysis	Mixed	spectromorphology (Smalley 1986, 1997; Thoresen & Hedman 2007; Blackburn 2011); structuring spectra (Howe 2023)
Temporal semiotic units	Mixed	Frey et al. 2009
Memetic analysis	Mixed	Dawkins 1989; Adkins 2009

We have also identified four sets inside the scheme—that is, four groupings of the analysis methodologies—: Three of the sets aggregate analysis methodologies that can be strongly argued to belong to mostly one of the semiotic levels much more than to the other two. But there are several methodologies that indeed capture some aspects of the three levels, and consequently they have been positioned in a more centred location. Their position in the overall diagram still represents the degree of closeness to each of the semiotic axes. Accordingly, the survey that follows is organized along these axes: poietic analyses (Section 4), neutral-level analyses (Section 5), aesthetic analyses (Section 6), and mixed approaches that draw on more than one level (Section 7).

The remainder of the article examines each of these methodologies in turn, explaining the reasoning behind its placement within the triangular scheme.

4. POIETIC ANALYSES

Nowakowski, M., P. Couprie, and J. Bresson. 2019. “Towards Deep Learning Strategies for Transcribing Electroacoustic Music.” In *Computer Music Modeling and Retrieval*, 28–37. Springer. https://doi.org/10.1007/978-3-030-70210-6_3.

Poietic analysis can be defined as the analysis of the composition process, and, in contrast to aesthetic analysis, tries to avoid the inevitable individuality involved in the listening process (Nucibella, Porcelluzzi, and Zattra 2005).

The growing prevalence of collaborative and co-creative practices—between composers, performers and technologists—further complicates the poietic account, since the creative process is distributed across several agents.

Program notes, sketches and composers' own commentaries constitute additional primary sources for poietic analysis.

4.1. Analysis of the creative process

Generative analysis is probably the most poietic of all analysis strategies, and usually is a multidisciplinary approach that combines musicology with computer science and perception. It studies the compositional process by the recuperation of the different material used during the creation of a work, or uses computer data as objective material to be analysed (Zattra 2005). Although this type of analysis is introduced as "Genetic analysis" by Zattra, we think that a more appropriate denomination is "analysis of the creative process". A very good example of this kind of analysis is the one done by Lorrain (1980) with respect to Risset's work *Inharmonique*, where the MUSIC IV-based score is examined in detail, trying to decipher as much as possible the composer's intentions.

A generative approach allows for an understanding of the structure and the sonority and temporal development within the piece. An evaluation of generative data permits, for example, the knowledge of the ways the computer or electronic equipment was used in the compositional process.

Some problems are recognizable in this type of analysis: first of all, we find that in many cases this type of analysis is not applicable if only the recording is available, so that means that the access to the material used during the creation of the work is fundamental for this type of analysis. Another problem is that technical means and sound results are not necessarily linked in a direct or evident way, making the method very difficult to apply in this regard.

This kind of analysis extends to the realm of computer-program-based generative art, where the music is generated with the minimum amount of human intervention (Collins 2008). These kinds of artworks are nowadays very common and they are very challenging for the analyst. As Young (2016) states: "In the last 20 years or more there has been an explosion of music making in which computers are not just in evidence but can be acknowledged as co-progenitors".

A fundamental aim in analysing such works is to explore the relation between source and production. Generative analysis of automated compositions must deal not only with the social and historical context of each piece, but also with a probably significant necessary amount of reverse engineering (Young 2016) in order to understand their functioning and how their sound world is created, as the exact details of the original generative program may not be recoverable (Collins 2008).

4.2. Philological analysis

Philology is fundamentally concerned with the different versions and states in which a work may exist. Philology can be thought of as the study of the structure and historical development of languages. Some authors (Orio and Zattra 2007, Zattra 2007) propose to borrow concepts from the philology of music for the study of electroacoustic music. In their point of view, the analysis should start from the research field named "philology of author", which encompasses the activities of the composer, in the light of his/her interrelation with personal aesthetics and cultural and biographical context. In Zattra's words: "The composer's 'mental' text is fundamental. It represents the musical intention, and it is worth being analyzed if one considers it as a support for the writing of the musical thought. Obsolescence and preservation are crucial problems in the study of electroacoustic music. Therefore, mental texts (of composers, technicians, etc.) are important to the preservation and analysis of musical works" (Zattra 2007).

A philological approach operates on different levels. In a first step, philologists examine documents in their physical reality, studying and evaluating the support media. They study the linguistic aspect

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and the literary aspect of the documents. In this sense, philology more generally refers to everything that aids in the restoration and illustration of the past. In the same way, the analyst must first of all determine the “language” of the music, but must also reconstruct the source’s history, situating and precisely describing it. Generally speaking, a true scholarly apparatus is indispensable (Bossis 2006).

One of the most paradigmatic examples of the application of this kind of analysis is the one applied to John Chowning’s *Stria* by Zattra (2007), based on “the hypothesis that this composition exists in several versions”. Thus, the analysis follows the historical genesis of the assembling of the sections, tracing the various stages of the assembling process, from the very first synthesis of the piece to two different and coexisting four-channel and stereophonic versions. This is, more generally, how musicologists tend to proceed philologically: by following the time-progression of a work through its successive versions and states rather than treating it as a single fixed object. Studies of compositional sketches, and of scores reconstructed after the fact, offer further material for such genetic readings.

4.3. Analysis of the compositional technique

Some composers think about compositional techniques and music technology as more than mere means, sometimes they are the main objective. A complete and exhaustive aesthetic approach to electroacoustic music needs to undertake a critique of the technical processes of composing and a characterization of the technological working environment, what is known as the *techné* of electroacoustic music (Di Scipio 1995). An implicit assumption for this kind of approach is that the functional characteristics of the equipment that was available during the formative years of composers materially influence the ways in which they develop their compositional aesthetic (Manning 2006). In other words, central to the analysis is the nature of the *techné* involved in the creative process.

Orio and Zattra (2007) stress the urgency for the analyst and the sound engineer to collaborate in order to find new analytical methods deriving from the combination of their competencies. Zattra (2005) proposes that analytical approaches to electroacoustic music can be based on knowledge of the technological environment used to generate the music. This implies to develop a methodology that is capable of characterizing the technical processes and the designing tools that make up the compositional environment, models, representations and knowledge-level strategies (Di Scipio 1995).

This point of view has several implications. One is that there must be a dimension where it is possible to grasp features of the cognitive and aesthetic potential that is specific to the medium being used. Another is the inevitable redefinition of the notion of material, and the relationship of this redefined material to form. Finally, one more evident issue is that technology changes very fast and adapts new forms very quickly and that here are several technical means for achieving similar results, providing difficulties for an approach of this kind.

Relevant work made with this approach are Di Scipio’s analyses of several of Xenakis’ electroacoustic works (Di Scipio 1998) and the analysis of Stockhausen’s *Octophonie* (Clarke and Manning 2008). In a similar vein, Clarke (2014) combines technological investigation and software emulation in his analysis of Truax’s *Riverrun*, and Haworth (2015) reads sound-synthesis procedures themselves as texts, foregrounding the ontological politics of electroacoustic and computer music.

5. NEUTRAL LEVEL ANALYSES

As noted in Section 3, the label 'neutral level' is a convenient shorthand rather than a claim of objectivity: visual and computational representations are themselves the product of analytical choices. Perhaps the most important reference for this kind of analysis is the work of Roy (2003), that focuses on acousmatic works influenced by Schaefferian theory. According to Roy, the analysis at the neutral level represents a verifiable methodological point of reference between the two extremes of the poietical and aesthetical levels, and provides a holistic approach for the analysis, establishing a global portrait of the symbolic processes that operate inside a composition (Roy 2003, 199). It bears repeating that this 'neutral' object is never encountered raw: every sonogram, descriptor or segmentation already embodies a particular analytical choice—the analysis window, the descriptors selected, the segmentation criterion—so that what counts as the trace depends on the very methods the analyst brings to it. In this strong sense we never hear the piece 'as it is', but always through the representations and listening habits we impose on it (Harcus 2017; Pihel 2024; Nattiez 2019).

5.1. Visual analysis

Representation is important for musical analysis because this is a study of a time art, and humans need to write time down to capture the ephemeral moment and study it (Couprie 2016). Good general-purpose transcribers in traditional notation do not even exist, let alone for electroacoustic music, as this type of music often contains spectral changes and other timbral aspects that make traditional music notation inadequate. Visual representations of music can aid the understanding of the music by allowing one to appreciate characteristics that may have been ignored previously and that are not easy to grasp aurally (Helmuth 1996).

Many analysts have proposed visual representations or scorings of the music. However, it is important to differentiate scoring from notation. Scoring is a tool for creation and allows generative thoughts to find their points of reference, while notation allows for a precise transmission of musical content to the performer (Bossis 2006). Notation can aid the scoring process. A systematic attempt to extend notation to unconventional playing techniques and electroacoustic sound is offered by Dimpker (2013).

One of the most common resources for this type of analysis is the spectrogram or sonogram (Waters and Ungvary 1990), which is widely used as a representation of the evolution of the spectrum in time. This representation allows the possibility to view a complete musical formal structure, to distinguish spectral tendencies, and to relate different levels of the structural hierarchy in a single scan.

Helmuth (1996) also proposes the sonogram as a useful representation of the frequency spectra and their evolutions in time but with the addition of several more layers containing relevant musical information, such as pitch relationships, phrasing, overall dynamic levels and processing techniques. Here, the sonogram is explored as a tool for understanding the sonic design of musical works. The additional layers can be included to clarify various aspects of the music and also other aspects such as pitch notation, phrase analysis, amplitude lines and a text line of comments are usually combined to form more accurate visual representations of music. A particularly interesting extension of sonograms are timbregrams (Park 2016), timbre space visualizations that change as a function of time. Whereas a spectrogram maps the raw distribution of acoustic energy across frequency and time, a timbregram instead traces the trajectory of a sound through a multidimensional timbre space derived from perceptual audio descriptors: the same passage appears not as a pattern of partials but as a path among timbral qualities such as brightness, so that features that are perceptually salient yet visually inconspicuous in a sonogram become legible over time.

Peeters, La Burthe, and Rodet (2002) have proposed an analysis method based on signal analysis that allows the generation of a visual representation of an audio recording based on self-similarity through time. The general strategy is to consider the audio signal as a succession of states at different scales, which correspond to the structure of a musical work, also at different scales. This approach is mostly valid when the music contains a great amount of repetition. Other transcription- and visualisation-based methods include the rhythmograms applied by Hirst (2014) to Normandeau's Onomatopoeias, the aural-transcription method of Mazzoli (2014, 2017), the comparative analysis of Couprie as applied by Cassaro (2018), and Sargenti's (2017) performance analysis of Stockhausen's Solo.

A very useful software for visual analysis is the *Acousmographie*, a program for the annotation of electronic music designed by GRM (Geslin and Lefevre 2004). This program combines background layers of waveform and spectrum with layers of marks and symbols, which allows the annotation of personal assessments, timeline marks and draws over the background layers. Another important tool is *EAnalysis* (Couprie 2016), a workspace where the analyst can create representations, import data from other software or recorded during performance, and analyse them. These kinds of systems provide a combination of a waveform representation, a sonogram and other layers that can be indexed and graphically annotated by the user. More recent open environments continue this lineage: Sonic Visualiser, for instance, combines waveform, sonogram and audio-descriptor layers that the analyst can annotate, while feature-extraction frameworks such as those developed by Lartillot make a broad palette of descriptors available for annotation and visualisation.

One clear advantage of this type of analysis is, that it is relatively easy to read for the analyst. However, it is well known that visual perception and auditory perception are not the same, they operate at very different resolutions scales. The sonogram "hears" the music as a whole mass of different spectra and amplitudes, while the human perception tends to recognize and categorize the different events and sound objects in the music discourse. In this sense, it can be argued that a sonogram is not a representation of the music perceived by the human ear, it is somehow too objective (Smalley 1997). Also, on any FFT-based visual representation of musical signal, there are problems related to the frequency and time scales, resolution and approximation. In practice, a single analysis window cannot resolve fast transients and low-frequency detail at once; linear-frequency sonograms misrepresent the ear's roughly logarithmic pitch perception; and perceptually salient masking relations are not displayed at all.

5.2. Computational Analysis

Under this category we consider several analytical approaches that take advantage of computer technology in order to gain insight and gather musical data that is either impossible or too hard to obtain by humans. This is not to be confused with analysis of the technique, which is the analysis of all technical means used in the composition of a piece of music, a methodology that is considered poietic in this article. Computational analysis exploits the ability of a computer to analyse sounds and recognize patterns within a musical piece in a meaningful manner for the analyst (Zattra 2005), and it does not necessarily involve a visualization stage. For this reason, it is considered a separate method from visual analysis. This type of analysis is essentially descriptive, a characterisation that applies equally to the visual analysis discussed in Section 5.1, as it provides mostly quantitative or measurable information from the object being studied, and as such it targets specifically the neutral level.

The recent progress accomplished in the domains of signal analysis and extraction of acoustic and formal parameters using powerful methods is remarkable (Bossis 2006). Automated analysis and matching tools can provide the musicologist with useful tools for the analysis and the philological study of electroacoustic music (Orio and Zattra 2007). This type of analysis has been strongly based on music information retrieval (MIR) research, which studies aural patterns and stream formation through the analysis of the audio signal, using methods derived from the field of machine

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listening. Some examples of current available tools are the Electro-Acoustic Music Analysis System (EASY) Toolbox and an accompanying methodology called Systematic and Quantitative Electro-Acoustic Music Analysis (SQEMA) (Park 2016). Other computational and exploratory frameworks include Dahan's temporal approach to electroacoustic music (Dahan 2011, 2020) and Miranda's (2010) proposal for a neurotechnology-based protocol of analysis.

There are several different applications for computational analysis. We can cite the *morphological sound descriptor* tool, based on the morphological theory of sounds objects by Pierre Schaeffer (Ricard and Herrera 2004), that can extract perceptual labels from sounds with good performance, avoiding a much more time-consuming manual labelling task. Computer-based multidisciplinary analysis (Nucibella, Porcelluzzi, and Zattra 2005) combines several features of computational listening to the analysis. Clarke (2012) has developed computer-based analytical tools that allows to recall specific passages from a work and relocate or juxtapose them, creating a very effective interactive aural analysis, which latest instantiation is named TIAALS, a set of generic software tools designed to facilitate an interactive aural approach to the analysis of electroacoustic music (Clarke, Dufeu, and Manning 2013).

It must be considered that while certain methods such as the FFT or automatic segmentation allow for a certain clarification of the structure of musical textures, they still remain considerably low the level of precision obtained by the careful reading of a traditional score (Bossis 2006).

In addition, applications that are based on periodicity do not apply very well for music that is essentially non-repeating or very complex. Electronic sounds have heterogeneous properties and are very different to regular instrumental harmonic sounds, and of course this type of analysis is limited by computational power.

Since the mid-2010s, computational analysis has been transformed by the wider availability of audio-descriptor toolkits and, more recently, by machine learning. Antunes, Malt and Manzolli (2019) propose a computer-based framework that characterises continuous and discontinuous textural works through psychoacoustic descriptors such as brightness, roughness and spectral flux, while Rossetti, Manzolli and Menezes (2018, 2019) track emergent timbre in mixed and live-electronic works using descriptors of spectral centroid, spectral flux and loudness. Such descriptors are now routinely extracted with open tools—Sonic Visualiser together with its Vamp plug-ins (including LibXtract), the Essentia library, and Couprie's EAnalysis, which integrates graphical transcription with descriptor analysis. A more recent development is the application of deep learning: Nowakowski, Couprie and Bresson (2019) explore neural strategies for the automatic transcription of electroacoustic music, and Onwuegbuna (2019) frames such work as a man-machine collaboration in which computation augments, rather than replaces, the interpretive work of the analyst. These approaches remain limited by the diversity of electroacoustic styles and by the scarcity of large labelled datasets, so that, for the foreseeable future, machine learning is best understood as an assistive layer within an essentially descriptive, neutral-level analysis.

6. AESTHESIC ANALYSES

There have been many approaches for aesthetic analysis. Du Preez (2022) published a selected chronologie of different proposals for analysis based on listening, ranging from Schaeffer's tipomorphologies in 1996 to the four part question and template for analysis proposed by Emmerson and Landy in 2016.

6.1 Analysis based on listening strategies

Delalande (1998) presents an explicit critique of Smalley's spectromorphology. He affirms that there is one type of analysis which would be unacceptable: one which aims to demonstrate what

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the music is, or what the form or the structure is. One must dismiss the illusion of the definitive unique analysis.

Anyone who has tried to analyse electroacoustic music (or even a slightly complex sound) knows very well that one can reveal an infinity of morphological characteristics and therefore configurations of morphologies. This general remark applies not only to every piece of music but also to every object. Whether explicitly stated or not, an analysis always implies a reduction. Only certain characteristics and configurations are selected from among all those available. The question of the objective of the analysis and the pertinence of characteristics in relation to this objective is thereby evaded.

Delalande proposes the opposite course of action. The objective of music analysis is to bring to light configurations which either reflect the choices (implicit or explicit) and actions of the composer, or which are needed to explain the reception behaviors of listeners, or both at once.

Once an analytical objective is chosen, it becomes possible to define a criterion of pertinence. On a general and theoretical level, there is no reason why pertinent units should coincide with morphological units. The best example comes from language where the phoneme, a functionally pertinent unit, is generally not a morphological unit and where, conversely, the syllable, a morphological unit as well.

Delalande proposes three different listening approaches (reception behaviors): taxonomic, emphatic and figurativisation.

As described in Smalley (1992), Pierre Schaeffer identified four modes of listening. Modes one and two are related to everyday listening, while modes three and four are specific to music listening. The modes are:

- Écouter. In this mode we are occupied with the provenance of sound and the message it carries. Our attention focuses on the occurrence or event attached to the sound. For example, if we hear the sound of a car travelling across the street we are not interested in the sound itself, but rather on what that sound means: maybe the car is running too fast and will hit us. This mode treats the sound as a sign of its source.
- Ouir. There is no intention to listen, but a sound reaches us and we cannot avoid it. An example of this could be a sudden explosion or a cry. This is the crudest level, the most elementary of perception.
- Entendre. This is a selective process where some sounds are preferred over others. According to its etymology, it means showing an intention to listen.
- Comprendre. This is a step further from the third mode. It means grasping a meaning, values, by treating the sound like a sign, referring to this meaning as a function of a language or a code (semantic hearing).

In Schaefferian theory, reduced listening is the attitude which consists in listening to the sound for its own sake, as a sound object by removing its real or supposed source and the meaning it may convey. In reduced listening our listening intention targets the event which the sound object is itself (and not to which it refers) and the values which it carries in itself (and not the ones it suggests).

In normal listening the sound is always treated as a vehicle. Reduced listening is therefore an anti-natural process, which goes against all conditioning. The act of removing all our habitual references in listening is a voluntary and artificial act which allows us to clarify many phenomena implicit in our perception.

The name reduced listening refers to the notion of phenomenological reduction because it consists to some extent of stripping the perception of sound of everything that is not itself in order to hear only the sound, in its materiality, its substance, its perceivable dimensions.

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Reduced listening and the sound object are thus correlates of each other; they define each other mutually and respectively as perceptual activity and object of perception.

Delalande does a taxonomic, emphatic and figurative analysis of the same piece, and the result of these different approaches are totally different, but complementary. Each behavior gives its particular form to the sound object. These different behaviors act like different analytical points of view.

Smalley (1992) extended Schaefferian modes combining them with Schachtel's autocentricity and allocentricity. As a result, he proposes the following relationships:

- Indicative. Corresponds to Schaeffer's mode one and considers sound as a message and object-centered. In this case, sound acts as a signal for something that is not sound in nature, but something needed, for example, to survive.
- Reflexive. Subject-centered, clearly auto-centric and is concerned with basic emotional responses to the object of perception. Object and emotion are not separable in this case.
- Interactive. Corresponds to Schaeffer's modes three and four and is clearly allocentric. It involves an active relationship on the part of the subject in continuously exploring the qualities and structure of the object. This relationship embraces structural hearing, aesthetic attitudes towards music and sounds, and Schaeffer's reduced listening.

The indicative and reflexive attitudes are the most dominant. It requires effort and willingness to engage in a more interactive relationship. Despite this, in our opinion, such a relationship is mandatory to really appreciate a piece of music and is the basic requisite to analyze it. Analyst listening (related to neutral level analysis) is not the same as the real-time listener (Roy 2003). Recent listener-oriented studies extend this perspective: Stead (2009) combines empirical enquiry with a post-structural analysis of intention and reception, Garrison (2013) reconsiders the figure of the model listener, and Wilson (2023) compares modes of listening in interpretation; the analysis of electroacoustic improvisation has likewise been addressed by Couprie (2022) and Hyvärinen (2022).

6.2. Analysis of the soundscape

A very important trend in electroacoustic music, first proposed by R. Murray Schafer in the sixties, encompasses the concept of a sonic landscape or more succinctly, a soundscape. This idea is related to what is called by Böhme-Mehmer (2008) a sounding souvenir, a construct analogous to a photograph in the visual domain. This kind of composition is a form of electroacoustic music characterized by the presence of recognizable environmental sounds and contexts, the purpose being to invoke the listener's associations, memories, and imaginations in relation to the soundscape (Truax 2008).

Soundscape composition is different from musique concrete or acousmatic music, as the original sounds must stay recognizable. Music created as soundscapes cannot be organised as instrumental music, as the composer's knowledge of the environmental and psychological context of the soundscape material is allowed to influence the shape of the composition at every level, producing the consequence that the composition is inseparable from reality. Ideally, soundscape composition should enhance our understanding of the world, and its influence carries over into everyday perceptual habits (Truax 2008).

Smalley (1992) elaborates on this by proposing what he calls surrogacy, as a means of describing the levels, or degrees, to which listeners perceptually relate sounds to real or imagined physical and gestural sources within an acousmatic listening situation. Surrogacy may describe the recognition of familiar instrumental or vocal production when reproduced in recorded form. As the degree of surrogacy becomes more remote, listeners may ascribe imagined causes to sounds and

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their spectro-morphological evolution, or apprehend the sound as abstract. Smalley terms these varying levels as orders of surrogacy. Cádiz, Fugellie and Eisner-Sagüés (2026) draw on these orders of surrogacy to examine how Chilean electroacoustic works mobilise recorded voice and testimony as vehicles of historical memory, proposing the notion of ‘acousmatic storytelling’.

Truax (2008) proposes that soundscape composition is much more an organising principle and not merely another style or sub-genre of electroacoustic music.

Wishart (1986: 43) proposes that the analysis of soundscapes should consider the source from which we imagine the sounds to come. In contra-position to Schaeffer’s concept of reduced listening, Wishart argues that listeners are more often aware of the source of a sound than the sound itself and that our mental apparatus is predisposed to allocate sounds to their sources. This is especially true when the composer purposefully utilizes recognizable sounds taken from the environment, as in soundscape composition.

What is the landscape of a piece of electroacoustic music? This is a difficult question and the answer is particular to that piece. Most people accustomed to concert music are disoriented when they attend a concert of electroacoustic music projected over loudspeakers. This disorientation is due to the inability to define an imaginable source, in the sense of a landscape, for the sounds perceived.

Smalley notices that the extensive range of sound sources admitted into music via the electroacoustic music medium has initiated a revolution in the sounding content of musical works, appealing to a variety of listening responses. (Smalley 1992)

What aspects of our perception of an aural image enter into the definition of an aural landscape? Wishart proposes that our perception of a landscape can be broken into three components: the nature of the perceived acoustic space, the disposition of sound-objects within the space and the recognition of individual sound objects (Wishart 1986: 45-49).

Unlike linguistic signs, which are essentially arbitrary, environmental sounds have specific aural qualities that are tied to their interpretation, and they cannot be changed arbitrarily without cognitive disruption (Truax 2008).

Truax (2008) proposes to invert the tradition of listening to environmental soundscapes as if they were music, and instead listening to electroacoustic music as if it were soundscape. This viewpoint certainly has important implications for the analysis, as it departs from spectro-morphology and the concept of sound objects, and provides a bridge between reduced listening and the various listening behaviors proposed by authors such as Delalande.

Basic soundscapes categories are hence useful for the analysis. Some perceptual habits, such as keynote sounds, signal sounds and soundmarks can be directly applied in the analysis. Keynote sounds provide background continuity, signals provide foreground information and they can become soundmarks if given uniqueness by cultural and social associations that transcend an immediate situation (Truax 2001).

6.3. Functional analysis

Another approach is the functional analysis outlined by Roy (2003). As stated by Roy himself “my functional approach is analytical and interpretative, and always based on listening” (Roy 1998). His technique employs a functional grid with symbols for forty-five functions classified into four main categories for orientation, stratification, process and rhetoric. The intention of such an approach is to comprehend units that are pertinent for the analysis. In his view, sound parameters interfere with one another too much in a complex synergy to be treated separately as far as electroacoustic music analysis is concerned (Roy 2003).

According to Gil De Morais (2015), for Roy, the “semiotic point of view adopted for functional analysis is the product of an inductive aesthetic that places the listener of an electroacoustic work in contact with its spontaneous perception. Thus, for him, the listener not only recognizes listened units from its morphological features, but also from the function they have and play in the work”. Even though Roy’s functional analysis is directed mostly towards acousmatic music, some authors have found it to be well suited for the analysis of mixed music (Gil De Morais 2015). Couprie, Narita, Hirano and Raboisson (2025) compare French and Japanese traditions in the study of acousmatic interpretation.

6.4. Cognitive Analysis

For Kendall (2016), the “key question to be answered by the analysis of electroacoustic music is: How does the listener’s experience of listening become meaningful?” Everyday listening is focused on events rather than on attributes of sounds (Gaver 1993). It is important to consider what listeners do in everyday life when making sense of the world around them (Kendall 2009). This question is far more significant for electroacoustic music than for traditional acoustic music because the expansive acoustic palette of electroacoustic music incorporates the sounds of the everyday world. Not just the sounds, but by immediate extension, the listening strategies, knowledge of auditory patterns and history of auditory associations that are inextricably linked to everyday listening.

Godoy (2006) suggests that the concept of reduced listening, typically used in analysis, is not in conflict with more fundamental principles of embodied cognition, the imagery of movement. Kendall (2014) proposes a model of five mental layers in order to clarify how the notion of layers contributes to our understanding of the listening experience. This model contains five different layers: Sensations, Gist, Locus, Context and Domains. Each layer can be described in terms of the potential meanings from within its own perspective, from which a dominant meaning will ultimately emerge.

An important example of this analysis is the one done by Kendall (2016), where he conducts the analysis of a short excerpt from Structure 16, a section of Stockhausen’s Elwell (2023) similarly organises an aural-space taxonomy for electroacoustic music grounded in causality and classification. *Telemusik*.

6.5. Narrative Analysis

Conceptions of narrative are central to time-based arts (Young 2016). Interactive narratives are inextricable from the way that we understand our encounters with digital technology. These encounters are formed into a narrative of episodic events, arranged and re-arranged at different levels of agency (Brown, Barker, and Del Favero 2011). The narratologist David Herman, as cited in Çamci (2016), defines narrative as “a basic human strategy for coming to terms with time, process, and change”. Every creative process develops its own type of narration, even independently of its creator’s purpose. Giomi and Ligabue (1998) propose to define a narrative-based analysis for electroacoustic music carried out through an aesthetic-cognitive methodology, mainly based on perception. This allows the examination of formal low-level signification units: starting from the level of the smallest elements (sound events) they proceed to the individuation of syntagmatic chains, compositional and signification strategies and to the segmentation of the structural and formal framework of the piece. This kind of analysis considers introductory and ending strategies, the use of analepsis (narrative repetition) and prolepsis (narrative anticipation), as well as semantic, metaphorical, musical and extra-musical psychological associations. The dialogue between musical structure and narrative semiotics has been developed more broadly by Grabócz (2019) in conversation with Greimas.

Çamci (2016) defines a gesture in electronic music as “an intentional narrative unit that coexists with other gestures in causal relationships at various timescales ... While sound-producing events in our environments also bear meaning, coexist at various timescales, and operate within causal networks, intentionality is what sets gestures apart as narrative units”. Although Çamci’s approach deals with both poietic and aesthetic dimensions, it is pertinent to consider that the notion of gesture can be utilised effectively as an analytical tool, particularly when dealing with the perceived narrative structure of the music.

Giannoukakis (2016) addressed the creation and analysis of media-based artworks by the classification and linking of multimedia morphologies, informed by cognitive linguistics, catastrophe theory and narratological studies. In the author’s words: “Examining narrative in terms of the media in which it is conveyed, requires, among other things, examining the semiotic value these media carry for narrative. Analysis of media as semiotic phenomena requires consideration of the codes and sensory channels that support them, leading to three broad media groupings: verbal, visual and aural. Narratological investigation, then, is concerned with the affordances and constraints these different types of groupings (and signs they carry) impose on narrative”.(pp)

Giomi and Ligabue (1998) applied this kind of strategy to six pieces of music, finding that fragments or more extended processes with a narrative nature were perceptible in almost all of the compositions they analysed. Çamci (2016) conducted an experiment with eighty participants that listened to several electroacoustic compositions, producing a categorical analysis of mental associations they evoked, that was explained in terms of narratological theories. Savva (2021) develops a related dramatized-performative narrative method for both composing and analysing electroacoustic music.

6.6. Mimetic analysis

Closely related to mimesis is the notion of surrogacy introduced in Section 6.2: Smalley’s orders of surrogacy describe the degrees to which listeners relate sounds to real or imagined sources, and mimetic discourse can be read as the compositional counterpart of these listening relationships.

Fischman (2008) aims to provide a framework and terminology for the discussion, analysis and creation of music that articulates mimetic discourse and structure. In doing so, it also intends to enable parallel and complementary analytical approaches to those concerned with the aural and spatial aspects of musical articulation. It proposes the notion of multidimensional mimetic space, taking as its point of departure Emmerson’s Language Grid (Emmerson 1986), and constructing additional axes representing compositional continua, similar in function to those proposed by spectro-morphology (Smalley 1986), but applicable to mimetic material.

Emmerson first defined mimetic discourse as that which imitates nature (i.e. the physical world) or aspects of human culture not usually associated with musical material. He also applied this concept in the construction of a language grid in order to understand the relationship between syntax and electroacoustic materials; where the latter form a continuum spanning from aural to mimetic. Hunt (2020) extends these concerns to the voice, examining the role of spoken and sung text in electroacoustic composition.

7. MIXED-ANALYSIS

The analyses grouped here under the heading of mixed analysis combine features of more than one semiotic level, drawing simultaneously on poietic, neutral-level and aesthetic perspectives. They occupy the central region of the classification scheme and are discussed together for that reason.

7.1. Spectro-morphological analysis

The term spectro-morphology was coined in 1986 by Dennis Smalley (already encountered above in the discussions of motion and listening), and it refers to a descriptive tool based on human aural perception that addresses the interaction between sound spectra and the ways they change and are shaped through time (Smalley 1986). In other words; spectro-morphology in this wide sense is a frame of reference for the understanding of structural relations and sonic behaviour as the temporal flux of the music progresses. In this area, Smalley has extended Schaeffer's typomorphological ideas for describing and analysing aural experiences (Smalley 1986; Smalley 1997). The two parts of the term refers to sound spectra and shaping through time. The spectra cannot exist without the morphology: something has to be shaped and that something must have sonic content.

Smalley's conception of spectro-morphology is ample, and encompasses several dimensions and variables such as gesture, behaviour, spectra, motion, growth processes, texture, space, intrinsic-extrinsic threads and structural levels, processes and functions. Since its conception, Smalley's spectro-morphology has equipped listeners and practitioners of electroacoustic music with appropriate and relevant vocabulary to describe the sound-shapes, sensations and evocations associated with experiences of sound. Its influence has been far reaching, inciting approaches to analysis, notation, composition and education through its flexible functionality and accessible pool of vocabulary (Blackburn 2011).

However, the term spectro-morphology has also been used by Thoresen and Hedman (2007) in a more restricted sense, focused on the extension of Schaeffer's typomorphology. Thoresen's main contribution is the refinement of the spectro-morphological models, considering several more variations by making a wider palette of symbols as opposed to letters or verbal designations to represent the analysis.

We have decided to consider Schaeffer's typomorphology, Thoresen spectro-morphology and Smalley's spectral-typology and morphological archetypes as equivalents in terms of their goals and semiotic level. Consequently, we grouped these together under the general term spectro-morphology. In this paper, other aspects of Smalley's framework, such as the concepts of motion or structuring processes are discussed under other categories of analysis.

From an analytical point of view, Smalley's proposal is extremely important, because in the first thirty-five years of electroacoustic activity no comprehensive methodology was proposed for analysis, even though Smalley's paper does not provide a clear indication of an analytical objective in the formulation of spectro-morphology. Pierre Schaeffer's theory of morphological types or typomorphologies (sound objects) (see 2.4) was the most serious attempt (Schaeffer 1966). Spectro-morphology extends Schaeffer's theory to include structural hierarchies and other properties like motion and space.

However, it is important to notice that spectro-morphology is not merely an analytical technique itself, it is rather a descriptive tool based on aural perception. A spectro-morphological approach sets out spectral and morphological models and processes, and provides a framework for understanding structural relationships and behaviours as experienced in the temporal flux of the music (Smalley 1997). Nevertheless, we have chosen to position spectro-morphology in a centred

position, as some proposals that are strongly related to it, such as Thoresen and Hedman (2007) or Blackburn (2011) are more related to the poietic or trace levels.

Smalley's examination becomes a systematic re-formulation and enlargement of Schaeffer's affirmations, preserving some of their original characteristics and conferring generality on many aspects. The bond between the two approaches is evident, especially in the relationship between spectral typologies and morphologies. In Smalley's definition, however, these terms are more exhaustive; they can be applied to different musical contexts and include other categories such as that of motion (Camilleri 1993).

In terms of morphology, Wishart (1986) differentiates intrinsic versus imposed morphology: intrinsic morphology is the natural morphology of a sound and imposed morphology is when there is a continuous input of energy applied to the sound. It is safe to assume that any sound-event has an intrinsic and an imposed morphology (Wishart 1986, 57). Most sounds have an intrinsic morphology, one that is stable and given by the natural attack-decay envelope of the sounds which is a consequence of the physical properties of the sounding medium. However, as most physical systems would require a continual energy input for the sounds to be continuously produced, these kind of sounds in fact possess an imposed morphology. Intrinsic morphology does not tell much about the gesture conveyed by the sound, while imposed morphology provides information about the gestural structure of sounds.

One of the important uses of spectro-morphology is for the creation of a visual representation of either the compositional process (Blackburn 2011) or the analysis of the trace (Thoresen and Hedman 2007), providing several benefits for the analyst. However, a weak point of this approach is the non-consideration of the spatial dimension on the typo-morphological score, which is sometimes fundamental for the comprehension of sound behaviours, like in the analysis of Parmerud's *"Les Objets Obscurs"* (Thoresen 2009). A final consideration that we can add to this critic is the fact that some symbols used in the spectro-morphological analysis (even if they come from Schaeffer, Smalley, Thoresen or Blackburn) do not necessarily correlate as visual reference to what we are hearing, as it occurs for example with the symbol of a complex sound spectrum, perhaps a square it's not the best representation of a complex spectra. From a compositional angle, Howe (2023) describes how the spectra of electroacoustic works can be structured coherently.

7.2. Analysis based on temporal semiotic units

Schaeffer did not consider the temporal and evolving aspects of musical objects. These considerations led researchers and composers of the Laboratoire Musique et Informatique de Marseille (MIM) to develop a new approach for musical analysis intended for electroacoustic music and based upon music temporal structure (Frey et al. 2009).

This group at MIM developed the concept of Temporal Semiotic Units (TSUs), following which musical segments convey meaning through their dynamic organization in time. To define the different TSU categories, they used a methodology based on collective listening of many different musical pieces. During the listening sessions, these researchers attempted to identify the specific temporal forms that developed between two and several seconds and that would convey meaning. A total of 19 TSU categories were thereby identified and named using representative labels: Chute, Trajectoire inexorable, Contracté-étendu, Elan, Etirement, En Flottement, Sans direction par divergence d'information, Lourdeur, Freinage, Obsessionnel, Qui avance, Qui tourne, Qui veut démarrer, Sans direction par excès d'information, Suspension-interrogation, En suspension, Par vagues, Stationnaire and Sur l'erre. Some of these units are limited in time, while others are not.

Finally, a written description of each TSU was also established that included a global morphological description, a semantic description, and other relevant characteristics, such as energy, intensity, and tension. A related linguistic borrowing is proposed by Link (2018), who adapts immediate constituent analysis as a model for electroacoustic music theory.

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7.3. Memetic Analysis

In his book *The Selfish Gene*, Richard Dawkins (1989) develops the concept of memetics, conceived as a theory of mental content based on an analogy with Darwinian evolution. A meme, analogous to a gene, is essentially a "unit of culture", such as an idea, belief or pattern of behaviour. Memes can be thought of neuronally-encoded cultural information, and their phenotypic products (behaviour and artefacts) spread through a process of imitation from one individual's memory to another (Adkins 2009). A meme is "hosted" in one or more individual minds, and can reproduce itself from mind to mind. As a consequence, memetics considers what would otherwise be regarded as one individual influencing another to adopt a belief, as a meme reproducing itself. Some examples of memes are tunes, ideas, catch-phrases, clothes fashions, ways of making pots or of building arches (Dawkins 1989, 192).

In electroacoustic music, the use of referential or anecdotal sound clearly differentiates electroacoustic from purely acoustic music and implies additional criteria to enable an appropriate segmentation and perceptual grouping of memes (Adkins 2009).

Narmour's implication-realization model and its extension outlined above is one model for the segmentation of sonic material. Adkins emphasizes that whatever the method of segmentation employed, the most important factor in attributing memetic status is co-equality (Adkins 2009). Once the meme is identified there are numerous other methods for further classifying sound material, for example a spectro-morphological approach.

8. CONCLUSIONS

We have proposed an exhaustive survey and critical review of some important methodologies for analysis of electroacoustic music that have been proposed in the literature. This review could be of help in the establishment of an adequate analytical lexicon and provide some solid foundations for a future comprehensive methodology for the analysis of electroacoustic music.

Different methodologies and principles for the analysis of electroacoustic music have been studied and presented. The final objective is to propose a consistent and exhaustive methodology. Such a methodology should consider all aspects of the musical phenomena, and take into account the particularities of the multiple subgenres of electroacoustic music. Such a systematisation matters now, because published analyses remain scarce and the field still lacks a shared vocabulary; but it matters even more for the coming decade, as machine-listening and machine-learning tools make large-scale, semi-automatic analysis increasingly feasible. A classification that locates each method within the tripartite is, in this respect, not only a way of organising what already exists but also a framework for situating the analytical tools—and the questions—that are still to come.

We also propose a classification scheme based on the semiology of music, where each methodology is categorized according to their function in terms of the tripartite proposed by Molino and Nattiez (1990). Although the methodologies surveyed here are presented one by one, this scheme is intended as the organising contribution of the article: the review itself is structured along the poietic (Section 4), neutral-level (Section 5), aesthetic (Section 6) and mixed (Section 7) axes, so that every method is located by its relation to the three semiotic levels. We hope that this classification scheme could be adopted towards an establishment of an analytical lexicon as desired by Camilleri (1993).

9. REFERENCES

Cádiz, R. et al 2026. *Analytical methodologies for electroacoustic music: a semiotic classification scheme and critical survey (pre-print draft)*

- Adkins, M. 2009. "The Application of Memetic Analysis to Electroacoustic Music." *Sonic Ideas* 1 (2): 34–41. <https://montyadkins.files.wordpress.com/2013/06/memetic-analysis-in-electronic-music.pdf>.
- Aguilar, Ananay. 2006. "El Papel de Los Procesos de Estructuración En El Análisis de Música Electroacústica." *Electroacoustic Music Studies Network (EMS)-MusicAcoustica*.
- Akkermann, Miriam. 2025. "Then and Now – Musical Needs and Intelligent Tools." In *Proceedings of the 50th International Computer Music Conference*, 212–215. Boston, MA: International Computer Music Association.
- Alvear, M., J. Meléndez, and R. Torres. 2024. "Semiótica triádica de Charles S. Peirce y creatividad en la interpretación de música experimental." *Música Hodie* 24. doi:10.5216/mh.v24.78403.
- Antunes, J., M. Malt, and M. Manzolli. 2019. "A Computer-Based Framework to Analyze Continuous and Discontinuous Textural Works Using Psychoacoustics Audio Descriptors." In *Proceedings of the Brazilian Symposium on Computer Music*. <https://doi.org/10.5753/SBCM.2019.10415>.
- Barrett, Natasha. 2002. "Spatio-Musical Composition Strategies." *Organised Sound* 7 (3): 313–23. doi:10.1017/S1355771802003114.
- Battier, Marc, and Leigh Landy. 2004. "Electroacoustic Musics: A Century of Innovation Involving Sound and Technology — Resources, Discourse, Analytical Tools." *Organised Sound* 9 (1): 1–2.
- Blackburn, Manuella. 2011. "The Visual Sound-Shapes of Spectromorphology: An Illustrative Guide to Composition." *Organised Sound* 16 (01). Cambridge Univ Press: 5–13.
- Bossis, Bruno. 2006. "The Analysis of Electroacoustic Music: From Sources to Invariants." *Organised Sound* 11 (2). Cambridge Univ Press: 101.
- Bramantyo, A. 2025. "Cultural Expression and Semiotic Adaptation: Tarasti's Framework in the Study of Modern Music." *Rast Műzikoloji Dergisi* 13 (41). doi:10.12975/rastmd.20251341.
- Bregman, Albert S. 1990. *Auditory Scene Analysis: The Perceptual Organization of Sound*. MIT Press.
- Brown, Neil C.M., Timothy S. Barker, and Dennis Del Favero. 2011. "Performing Digital Aesthetics: The Framework for a Theory of the Formation of Interactive Narratives." *Leonardo* 44 (3): 212–19. doi:10.1162/LEON_a_00165.
- Cádiz, Rodrigo F., Daniela Fugellie, and Federico Eisner-Sagüés. 2026. "Chilean Electroacoustic Music as Living Memory: Reinterpreting History Through Sound." *Organised Sound* 31: 57–64. doi:10.1017/S1355771825100800.
- Çamci, Anil. 2016. "Imagining through Sound: An Experimental Analysis of Narrativity in Electronic Music." *Organised Sound* 21 (3): 179–91. doi:10.1017/S1355771816000169.
- Camilleri, Lelio. 1993. "Electro-Acoustic Music: Analysis and Listening Processes." *Sonus Contemporary Music Materials* 1 (3): 26.
- Camilleri, Lelio, and Denis Smalley. 1998. "The Analysis of Electroacoustic Music: Introduction." *Journal of New Music Research* 27 (1–2): 3–12. doi:10.1080/09298219808570737.
- Cassaro, Dilson Colman. 2018. "Comparative Analysis as a Tool to Identify Aspects of Electroacoustic Music Composers' Language." In *Proceedings of the 11th International Conference of Students of Systematic Musicology*.
- Clarke, Michael. 2012. "Analysing Electroacoustic Music: An Interactive Aural Approach." *Music Analysis* 31 (iii): 347–80. doi:10.1111/j.1468-2249.2012.00339.x.
- Clarke, Michael. 2014. "From Technological Investigation and Software Emulation to Music Analysis: An Integrated Approach to Barry Truax's Riverrun." In *Proceedings ICMC|SMC 2014*. Athens.
- Clarke, Michael, Frédéric Dufeu, and Peter Manning. 2013. "TIAALS: A New Generic Set of Tools for the Interactive Aural Analysis of Electroacoustic Music." *Proceedings of the Electroacoustic Music Studies Conference*. www.ems-network.org.
- Clarke, Michael, and Peter Manning. 2008. "The Influence of Technology on the Composition of Stockhausen's Octophonie, with Particular Reference to the Issues of Spatialisation in a Three-Dimensional Listening Environment." *Organised Sound* 13 (03). Cambridge University Press: 177. doi:10.1017/S1355771808000277.

Cádiz, R. et al 2026. *Analytical methodologies for electroacoustic music: a semiotic classification scheme and critical survey (pre-print draft)*

- Collins, Nick. 2008. "The Analysis of Generative Music Programs." *Organised Sound* 13 (03): 237. doi:10.1017/S1355771808000332.
- Cook, Nicholas. 1998. *Analysing Musical Multimedia*. Oxford: Oxford University Press.
- Couprie, Pierre. 2016. "EAnalysis: Developing a Sound-Based Music Analytical Tool." In *Expanding the Horizon of Electroacoustic Music Analysis*, edited by Simon Emmerson and Leigh Landy, 170–94. Cambridge University Press. doi:10.1017/CBO9781316339633.009.
- Couprie, Pierre. 2022. "Analytical Approaches to Electroacoustic Music Improvisation." *Organised Sound* 27 (2). doi:10.1017/S1355771821000571.
- Couprie, Pierre, Kazuko Narita, Saburo Hirano, and Nathanaëlle Raboisson. 2025. "The Relationship Between France and Japan in Studying the Interpretation of Acousmatic Music." In *Proceedings of the 50th International Computer Music Conference*, 200–207. Boston, MA: International Computer Music Association.
- Curry, Ben. 2017. "Valency–Actuality–Meaning: A Peircean Semiotic Approach to Music." *Journal of the Royal Musical Association* 142 (2). doi:10.1080/02690403.2017.1361177.
- Dahan, Kevin. 2011. "Electroacoustic Music: Overcoming Analysis Paralysis." In *Proceedings of the International Computer Music Conference*. Huddersfield.
- Dahan, Kevin. 2020. "A Temporal Framework for Electroacoustic Music Exploration." *Organised Sound* 25 (2): 165–178. doi:10.1017/S1355771820000151.
- Dawkins, R. 1989. *The Selfish Gene*. 2nd editio. Oxford: Oxford University Press.
- Delalande, François. 1998. "Music Analysis and Reception Behaviours - Sommeil by Pierre Henry." *Journal of New Music Research* 27 (1–2): 13–66. doi:10.1080/09298219808570738.
- Di Scipio, Agostino. 1995. "Centrality of Techné for an Aesthetic Approach on Electroacoustic Music." *Journal of New Music Research* 24 (4): 369–83. <http://ears.pierrecouprie.fr/spip.php?article539>.
- . 1998. "Compositional Models in Xenakis's Electroacoustic Music." *Perspectives of New Music* 36 (2): 201–43. doi:10.2307/833529.
- Dimpker, Christian. 2013. *Extended Notation: The Depiction of the Unconventional*. Musikwissenschaft 17. Berlin: LIT Verlag.
- du Preez, Charles. 2022. *Listener-Based Analysis of Electroacoustic Music: A Selected Chronology of Methods as Applied to Amprorification #1 and Sorstalanság*. PhD diss., University of Arizona.
- Elwell, Kramer. 2023. *Organizing an Aural Space Taxonomy for Electroacoustic Music: Analysis via Causality and Classification*. PhD diss., University of California, Santa Barbara.
- Emmerson, S, and L Landy. 2016. *Expanding the Horizon of Electroacoustic Music Analysis*. Edited by Simon Emmerson and Leigh Landy. Cambridge: Cambridge University Press. doi:10.1017/CBO9781316339633.
- Fischman, Rajmil. 2008. "Mimetic Space Unravelling." *Organised Sound* 13 (02). doi:10.1017/S1355771808000150.
- Frey, Aline, Céline Marie, Lucie Prod'Homme, Martine Timsit-Berthier, Daniele Schön, and Mireille Besson. 2009. "Temporal Semiotic Units as Minimal Meaningful Units in Music? An Electrophysiological Approach." *Music Perception: An Interdisciplinary Journal* 26 (3). University of California Press Journals: 247–56. doi:10.1525/mp.2009.26.3.247.
- Garrison, Travis. 2013. *Reconsidering the Model Listener: An Exploration in the Critical Analysis of Electroacoustic Music*. PhD diss., University of Florida.
- Gatt, Michael. 2015. "The OREMA Project: A Call for the Liberation of Sound Analysis." *Organised Sound* 20 (3): 316–22. doi:10.1017/S1355771815000242.
- . 2016. "OREMA an Analytical Community." In *Expanding the Horizon of Electroacoustic Music Analysis*, edited by Simon Emmerson and Leigh Landy, 148–69. Cambridge University Press.
- Gayou, Evelyne. 2006. "Analysing and Transcribing Electroacoustic Music: The Experience of the Portraits Polychromes of GRM." *Organised Sound* 11 (2): 125–29. doi:10.1017/S1355771806001385.
- Geslin, Yann, and Adrien Lefevre. 2004. "Sound and Musical Representation: The Acousmographie Software." In *Proceedings of the International Computer Music*
- Cádiz, R. et al 2026. *Analytical methodologies for electroacoustic music: a semiotic classification scheme and critical survey (pre-print draft)*

Conference.

- Giacosa, G. 2023. "Musical Meaning and the Semiotic Hierarchy: Towards a Cognitive Semiotics of Music." *The Public Journal of Semiotics* 10. doi:10.37693/pjos.2023.10.25325.
- Giannoukakis, Marinos. 2016. "Narrative in Form: A Topological Study of Meaning in Transmedial Narratives." *Organised Sound* 21 (3): 260–72. doi:10.1017/S1355771816000236.
- Gil De Moraes, Ronan. 2015. "Listening and Analysis of Mixed Electroacoustic Music as Tools for Interpretation's Construction." *International Journal of Music and Performing Arts* 3 (2): 7–18. doi:10.15640/ijmpa.v3n2a2.
- Giomi, Francesco, and Marco Ligabue. 1998. "Understanding Electroacoustic Music: Analysis of Narrative Strategies in Six Early Compositions." *Organised Sound* 3 (01). Cambridge Univ Press: 45–49.
- Godoy, Rolf Inge. 2006. "Gestural-Sonorous Objects: Embodied Extensions of Schaeffer's Conceptual Apparatus." *Organised Sound* 11 (2). Cambridge University Press: 149–57. doi:10.1017/S1355771806001439.
- Grabócz, Márta. 2019. "Structure et sens en musique: dialogue avec Greimas." *Langages* 213. doi:10.3917/LANG.213.0079.
- Harcus, A. 2017. *The Varieties of Tone Presence: On the Meanings of Musical Tone in Twentieth-Century Music*. PhD dissertation.
- Haworth, Christopher. 2015. "Sound Synthesis Procedures as Texts: An Ontological Politics in Electroacoustic and Computer Music." *Computer Music Journal* 39 (1). doi:10.1162/COMJ_a_00284.
- Helmuth, Mara. 1996. "Multidimensional Representation of Electroacoustic Music." *Journal of New Music Research* 25 (1). Taylor & Francis Group: 77–103. doi:10.1080/09298219608570698.
- Hirst, David. 2014. "The Use of Rhythmograms in the Analysis of Electroacoustic Music, with Application to Normandeau's Onomatopoeias Cycle." In *Proceedings ICMC|SMC 2014*. Athens.
- Howe, Hubert. 2023. "Structuring Spectra in Electroacoustic Music." *Organised Sound* 28 (3). doi:10.1017/S1355771823000535.
- Hunt, Edmund. 2020. "Composition as Commentary: Voice and Poetry in Electroacoustic Music." *Journal for Artistic Research*.
- Hyvärinen, Lauri. 2022. "Gesture and Texture in the Electroacoustic Improvised Music of Jin Sangtae, Hong Chulki and Tetuzi Akiyama." *Organised Sound* 27 (2).
- Ikeshiro, Ryo, and PerMagnus Lindborg. 2021. "SoundLab and Electroacoustic Music in Hong Kong." *Emille* 19.
- Kaiser, Jeff. 2025. "Developing a Pedagogy and Framework for the Analysis of Contemporary Electronic Music." In *Proceedings of the 50th International Computer Music Conference*, 316–321. Boston, MA: International Computer Music Association.
- Kendall, Gary S. 2006. "Juxtaposition and Non-Motion: Varese Bridges Early Modernism to Electroacoustic Music." *Organised Sound* 11 (2): 159–71. doi:10.1017/S1355771806001440.
- . 2010. "Spatial Perception and Cognition in Multichannel Audio for Electroacoustic Music." *Organised Sound* 15 (3). Lane Medical Library / Stanford University Medical Center: 228–38. doi:10.1017/S1355771810000336.
- . 2014. "The Feeling Blend: Feeling and Emotion in Electroacoustic Art." *Organised Sound* 19 (2): 192–202. doi:10.1017/S1355771814000132.
- Kendall, Gary S., and Mauricio Ardila. 2007. "The Artistic Play of Spatial Organization: Spatial Attributes, Scene Analysis and Auditory Spatial Schemata." In *Computer Music Modeling and Retrieval. Sense of Sounds*, 125–38. Berlin, Heidelberg: Springer Berlin Heidelberg. doi:10.1007/978-3-540-85035-9_8.
- Kendall, Gary S. 2008. "What Is an Event? The Event Schema, Circumstances, Metaphor and Gist." *Proceedings of the 2008 International Computer Music Conference*. [https://pure.qub.ac.uk/portal/en/publications/what-is-an-event-the-event-schema-circumstances-metaphor-and-gist\(0f34769b-b3d7-4bba-9c75-bd9e061bc92a\).html](https://pure.qub.ac.uk/portal/en/publications/what-is-an-event-the-event-schema-circumstances-metaphor-and-gist(0f34769b-b3d7-4bba-9c75-bd9e061bc92a).html).
- . 2016. "Listening and Meaning: How a Model of Mental Layers Informs Electroacoustic
- Cádiz, R. et al 2026. *Analytical methodologies for electroacoustic music: a semiotic classification scheme and critical survey (pre-print draft)*

- Music Analysis." In *Expanding the Horizon of Electroacoustic Music Analysis*, 31–57. Cambridge: Cambridge University Press. doi:10.1017/CBO9781316339633.003.
- Lalitte, Philippe. 2006. "Towards a Semiotic Model of Mixed Music Analysis." *Organised Sound* 11 (02): 93. doi:10.1017/S1355771806001348.
- Landy, Leigh. 2007. *Understanding the Art of Sound Organization*. MIT Press.
- Licata, Thomas. 2002. *Electroacoustic Music. Analytical Perspectives*. Wesport, Connecticut: Greenwood Press.
- Link, Martin Wilhelm Cornelis. 2018. "Immediate Constituent Analysis as a Future Model of Electroacoustic Music Theory." Conference paper.
- Lorrain, D. 1980. *Inharmonique: Analyse de La Bande Magnetique de l'oeuvre de Jean-Claude Risset*. Rapports IRCAM. Centre Georges Pompidou. <https://books.google.com/books?id=XANAAQAIAAJ>.
- Mahnkopf, Claus-Steffen, Frank Cox, and Wolfram Schurig, eds. 2006. *Critical Composition Today*. New Music and Aesthetics in the 21st Century 5. Hofheim: Wolke Verlag.
- Manning, Peter. 2006. "The Significance of Techne in Understanding the Art and Practice of Electroacoustic Composition." *Organised Sound*. doi:10.1017/S1355771806000112.
- Mazzoli, Mario. 2014. *Emerging Musical Structures: A Method for the Transcription and Analysis of Electroacoustic Music*. PhD diss., City University of New York.
- Mazzoli, Mario. 2017. "Emerging Musical Structures in Electroacoustic Music." In *Proceedings of the 9th European Music Analysis Conference (EuroMAC 9)*, Strasbourg.
- Merlier, Bertrand. 2018. "Space Notation in Electroacoustic Music: From Gesture to Signs." In *Proceedings of the International Conference on Technologies for Music Notation and Representation (TENOR)*.
- Miranda, Eduardo Reck. 2010. "New Protocols in Electroacoustic Music Analysis: A Neurotechnology Approach to the Analysis of Electroacoustic Music: A Proposition."
- Morthenson, J. W. 1985. "The Concept of Meaning in Electronic Music." In *Proceedings of the ICM Conference on Electro-Acoustic Music*.
- Nattiez, Jean-Jacques. 1990. *Music and Discourse: Toward a Semiology of Music*. Princeton University Press. doi:10.2307/431984.
- Nattiez, Jean-Jacques. 2019. "Lecture des lectures de mon travail." doi:10.7202/1062436AR.
- Noble, Jason, Tanor Borin, and Stephen McAdams. 2020. "Experiences of Time and Timelessness in Electroacoustic Music." *Organised Sound* 25 (2). doi:10.1017/S135577182000014X.
- Norris, M. 1999. "Reinstating Interpretation: The Status of Analysis in an Electroacoustic Context." In *Australasian Computer Music Conference*. Victoria: University of Wellington.
- Nucibella, Francesca, Savino Porcelluzzi, and Laura Zattra. 2005. "Computer Music Analysis via a Multidisciplinary Approach." In *Proceedings of the Sound and Music Computing Conference*. <http://smc.afim-asso.org/smc05/papers/FrancescaNucibella/FNSPLZ.pdf>.
- Onwuegbuna, E. 2019. "Digianalysis: The Man-Machine Collaboration in Music Analysis." In *Artificial Intelligence - Applications in Medicine and Biology*. IntechOpen. <https://doi.org/10.5772/INTECHOPEN.84355>.
- Orio, Nicola, and Laura Zattra. 2007. "Audio Matching for the Philological Analysis of Electroacoustic Music." In *Proceedings of the 2007 International Computer Music Conference*, 157–64. Copenhagen.
- Otondo, Felipe. 2008. "Contemporary Trends in the Use of Space in Electroacoustic Music." *Organised Sound* 13 (1): 77–81. doi:10.1017/S1355771808000095.
- Park, Tae Hong. 2016. "Exploiting Computational Paradigms for Electroacoustic Music Analysis." In *Expanding the Horizon of Electroacoustic Music Analysis*, edited by Simon Emmerson and Leigh Landy, 123–47. Cambridge University Press. doi:10.1017/CBO9781316339633.007.
- Peeters, Geoffroy, Amaury La Burthe, and Xavier Rodet. 2002. "Toward Automatic Music Audio Summary Generation from Signal Analysis." In *ISMIR 2002, 3rd International Conference on Music Information Retrieval*. <http://ismir2002.ismir.net/proceedings/02-FP03-3.pdf>.
- Pihel, E. 2024. "Beyond Structure and Chance: Listener and Irreversible Time in Musical Signification." In *Theory and History in the Human and Social Sciences*. doi:10.1007/978-3-03-12-0000-0_12.
- Cádiz, R. et al 2026. *Analytical methodologies for electroacoustic music: a semiotic classification scheme and critical survey (pre-print draft)*

- Postiga, M., S. Carvalho, and J. Mota. 2025. "Do corpo ao som: o gesto decodificado pela análise musical." doi:10.6063/motricidade.41516.
- Reybrouck, Mark. 2017. "Music and Semiotics: An Experiential Approach to Musical Sense-Making." doi:10.5772/67860.
- Ricard, Julien, and Perfecto Herrera. 2004. "Morphological Sound Description: Computational Model and Usability Evaluation." Audio Engineering Society. <http://www.aes.org/e-lib/browse.cfm?elib=12717>.
- Risset, Jean-Claude. 2002. "Preface." In *Electroacoustic Music. Analytical Perspectives.*, edited by Thomas Licata, xii–xviii. Westport, Connecticut: Greenwood Press.
- Rossetti, D., M. Manzolli, and J. Menezes. 2018. "Emergent Timbre and Extended Techniques in Live-Electronic Music: An Analysis of Desdobramentos Do Contínuo Performed by Audio Descriptors." *Música Hodie* 18 (1). <https://doi.org/10.5216/MH.V18I1.53568>.
- Rossetti, D., M. Manzolli, and J. Menezes. 2019. "Analysis of Granular Acousmatic Music: Representation of Sound Flux and Emergence." *Organised Sound* 24 (2): 202–215. <https://doi.org/10.1017/S1355771819000244>.
- Roy, Stéphane. 2003. *L'analyse Des Musiques Électroacoustiques : Modèles et Propositions*. L'Harmattan. <http://www.editions-harmattan.fr/index.asp?navig=catalogue&obj=livre&isbn=9782296344310>.
- Roy, Stéphane. 1998. "Functional and Implicative Analysis of Ombres Blanches." *Journal of New Music Research* 27 (1–2): 165–84. doi:10.1080/09298219808570743.
- Sad, J. C. 2020. "Timbre y significación: por una semiología de las formas sonoro-simbólicas." doi:10.19179/2319-0868/750.
- Salynas, Marius. 2021. "Semiological Interpretation of the Electroacoustic Music Phenomenon." *Scientific Herald of Tchaikovsky National Music Academy of Ukraine* 132. doi:10.31318/2522-4190.2021.132.249964.
- Sargenti, Simonetta. 2017. "Analysis of Electroacoustic and Interactive Music Works: Solo by Karlheinz Stockhausen, an Example of Performance Analysis." In *Proceedings of the Electroacoustic Music Studies Network Conference*, Nagoya.
- SATMUS (Sociedad de Análisis y Teoría Musical). 2023–. *Súmula: Revista de Teoría y Análisis Musical*. <https://www.satmus.org/>.
- Savva, Dimitrios. 2021. *Dramatized-Performative Narrative: A Novel Method for Composing Electroacoustic Music*. PhD diss., University of Sheffield.
- Schaeffer, Pierre. 1966. *Traité des objets musicaux*. Paris: Éditions du Seuil.
- Simoni, Mary. 2006. *Analytical Methods of Electroacoustic Music*. New York, NY, USA: Routledge. <https://books.google.com/books?hl=en&lr=&id=loAYDQAAQBAJ&oi=fnd&pg=PP11&dq=Analytical+methods+of+electroacoustic+music&ots=yNypZVlh0&sig=Jpu1faC0tCZYbMswTj-L7rMT1eg>.
- Smalley, Denis. 1986. "Spectro-Morphology and Structural Processes." In *The Language of Electroacoustic Music*, edited by Simon Emmerson, 61–93. Harwood Academic.
- . 1997. "Spectromorphology : Explaining Sound-Shapes." *Organised Sound* 2 (2): 107–26.
- Stead, Michelle. 2009. "Examining Intention and Reception in Electroacoustic Music: An Empirical Enquiry and Post-structural Analysis." University of Western Sydney.
- Stockhausen, Karlheinz. 1998. "Five Revolutions Since 1950." In *Contemporary Composers on Contemporary Music*, edited by Eliot Schwartz and Childs Barney, 380–383. Da Capo Press.
- Tarasti, E. n.d. "Los signos en la historia de la música, historia de la semiótica musical." doi:10.35494/topsem.2008.1.19.167.
- Thoresen, Lasse. 2009. "Sound-Objects, Values and Characters in Åke Parmerud's Les Objets Obscurs, 3rd Section." *Organised Sound*. doi:10.1017/S1355771809990124.
- Thoresen, Lasse, and Andreas Hedman. 2007. "Spectromorphological Analysis of Sound Objects: An Adaptation of Pierre Schaeffer's Typomorphology." *Organised Sound* 12 (02). Cambridge University Press (CUP): 129. doi:10.1017/s1355771807001793.
- Truax, Barry. 2008. "Soundscape Composition as Global Music: Electroacoustic Music as
- Cádiz, R. et al 2026. *Analytical methodologies for electroacoustic music: a semiotic classification scheme and critical survey (pre-print draft)*

- Soundscape*." *Organised Sound* 13 (2): 103–9. doi:10.1017/S1355771808000149.
- Waters, Simon, and Tamas Ungvary. 1990. "The Sonogram: A Tool for Visual Documentation of Musical Structure." In *Proceedings of the International Computer Music Conference*, 159–62. Glasgow, United Kingdom. [https://pure.qub.ac.uk/portal/en/publications/the-sonogram-a-tool-for-visual-documentation-of-musical-structure\(ccfa006b-7770-4e29-9fda-b0f8ba39719c\).html](https://pure.qub.ac.uk/portal/en/publications/the-sonogram-a-tool-for-visual-documentation-of-musical-structure(ccfa006b-7770-4e29-9fda-b0f8ba39719c).html).
- Wilson, Lauren. 2023. "Modes of Listening in the Interpretation of Electroacoustic Music." In *Proceedings of The 21st Century Guitar Conference 2019 & 2021*, 333–348.
- Wishart, Trevor. 1986. "Sound Symbols and Landscapes." In *The Language of Electroacoustic Music*, edited by Simon Emmerson. New York: Harwood Academic.
- Young, John. 2016. "Forming Form." In *Expanding the Horizon of Electroacoustic Music Analysis*, edited by Simon Emmerson and Leigh Landy, 58–79. Cambridge University Press.
- Young, Michael. 2016. "Interactive and Generative Music: A Quagmire for the Musical Analyst." In *Expanding the Horizon of Electroacoustic Music Analysis*, edited by Simon Emmerson and Leigh Landy, 80–101. Cambridge University Press.
- Zattra, Laura. 2004. "Searching for Lost Data: Outlines of Aesthetic—Poietic Analysis." *Organised Sound* 9 (1): 35–46. doi:10.1017/S1355771804000068.
- . 2005. "Analysis and Analyses of Electroacoustic Music." *Spectrum* 1: 2005–6. doi:10.1126/science.1125017.
- . 2006. "The Identity of the Work: Agents and Processes of Electroacoustic Music." *Organised Sound* 11 (2): 113–18. doi:10.1017/S1355771806001361.
- . 2007. "The Assembling of Stria by John Chowning: A Philological Investigation." *Computer Music Journal* 31 (3). MIT Press: 38–64.